

Continuous Area Occupancy-Based Optimization of Cycle-Based Max Pressure Control in Mixed Traffic

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Abstract. ...

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1 Introduction

Traffic congestion remains a critical challenge for sustainable urban development, particularly in developing countries such as Vietnam, where mixed traffic flows are dominated by motorcycles that typically account for 85% to 90% of the total traffic volume [5,10]. As documented in the literature, the non-lane-based, free-filtering, and swarming behaviors of dense motorcycle streams render traditional car-centric signal control strategies highly ineffective [14]. Unlike motor vehicles that follow structured queuing patterns, motorcycles fill available road gaps continuously, causing traditional discrete vehicle-counting methods to produce systematic inaccuracies. This leads to functional limitations in existing fixed-time signal plans and a measurable degradation of actual intersection capacity [6].

Although deploying real-time adaptive signal control frameworks such as the Max Pressure (MP) architecture is increasingly urgent, direct field trials face substantial barriers owing to limited physical sensor infrastructure and the high risk of cascading gridlocks during controller calibration [12,15]. To bypass these practical constraints, a Digital Twin (DT) architecture offers a viable solution by providing a safe, high-fidelity virtual testing environment prior to real-world deployment [4,1]. By applying Agent-Based Modeling (ABM) on the open-source GAMA multi-agent simulation platform [11,8], this study constructs a virtual testbed designed to replicate complex mixed-traffic interactions and emergent swarming behaviors under controlled conditions.

For the control strategy, the Cycle-Based Max Pressure (C-MP) algorithm is selected due to its proven ability to maximize network throughput through decentralized queue balancing while satisfying industrial fixed-cycle constraints [3].

To overcome the vulnerability of vision-based pipelines to perspective distortion, vehicle occlusion, and minor camera displacements that misalign rigid Regions of Interest (ROIs), this study proposes replacing discrete vehicle counts with a continuous state-space metric termed Continuous Area Occupancy (CAO), drawing on spatial saturation principles adapted from aerial surveillance concepts [2]. Leveraging YOLOv8 semantic segmentation, the proposed framework isolates the drivable road surface dynamically, thereby providing accurate, infrastructure-resilient spatial pressure values as inputs to the C-MP controller [13].

The experimental methodology encompasses two core phases: offline computer vision processing for baseline validation, and large-scale virtual network evaluation within the GAMA-based Digital Twin. Through a comparative analysis of the proposed CAO-based C-MP, the traditional vehicle-counting C-MP, and standard fixed-time control, this study benchmarks average delay, queue length, and throughput. The results demonstrate the operational superiority and structural resilience of the proposed system under dense, motorcycle-dominated traffic conditions.

2 Literature Review

The field of network-level traffic signal control has evolved considerably, progressing from rigid, centralized heuristics toward dynamic, distributed feedback strategies. In particular, the Max Pressure (MP) control framework has attracted considerable research attention due to its mathematical capacity to stabilize traffic networks under high demand conditions [12]. Subsequent studies extended this foundational theory by incorporating localized storage constraints, boundary capacities, and turning movement dynamics to better accommodate varied network topologies [9]. To facilitate practical field deployment, the Cycle-Based Max Pressure (C-MP) framework was introduced, allocating green time splits proportionally at cycle boundaries and rendering the algorithm compatible with standard industrial signal controllers while preserving network stability [3]. In parallel, high-fidelity microscopic simulation platforms—notably the GAMA multi-agent simulation platform—have been widely adopted to replicate complex urban traffic dynamics and evaluate advanced control strategies in a controlled environment prior to field implementation [11, 8]. However, these conventional frameworks are generally bounded by the assumption of structured, lane-based environments and homogeneous vehicle profiles. In non-lane-based networks, the linear relationship between discrete vehicle counts and physical intersection pressure frequently deviates from actual conditions, limiting the ability to accurately reflect real urban traffic states.

Concurrently, the research community has increasingly addressed the complexities of mixed traffic streams common in developing nations, where highly dense, non-lane-based motorcycle traffic constitutes a dominant share of urban transport flows [5, 10]. Early approaches applied static Passenger Car Equivalents (PCE) or Passenger Car Units (PCU) to estimate road capacity and convert heterogeneous flows into a standardized metric [6]. While these conversion methods

provide a structured adjustment for vehicle size differences, they face inherent limitations in highly dynamic, space-filling environments. Unlike passenger cars, dense motorcycle streams frequently exhibit non-linear swarming behaviors and aggregate compactly within any available road space, substantially influencing overall intersection dynamics [14]. Consequently, traditional discrete counting methods may not adequately capture the continuous spatial saturation of such flows, leading to inaccurate pressure estimations and suboptimal phase allocations. To refine these estimations, recent works have explored integrating heuristics with microscopic simulation to optimize signalized intersections under motorcycle-dominated conditions [13].

On the technological side, the development of Digital Twins for intersection management has accelerated, offering virtual environments for real-time traffic monitoring, performance evaluation, and systematic controller calibration [4,1]. Within these frameworks, adaptive traffic control systems have frequently leveraged deep learning object detectors, including the YOLO model family, alongside Digital Twin architectures to automate data extraction within predefined geometric Regions of Interest (ROIs) [7]. Nonetheless, vision-based estimation pipelines encounter practical challenges under real-world infrastructure constraints. Urban cameras are often mounted at oblique angles that introduce perspective distortion and inter-vehicle occlusion. Additionally, environmental factors such as wind or structural vibrations can cause minor camera displacements, resulting in misalignment between the rigid geometric ROI and the physical road boundaries. Such misalignment can introduce systematic errors into the data pipeline, affecting the reliability of downstream controller state estimations [15].

To address these constraints, this study proposes an alternative continuous state-space formulation termed Continuous Area Occupancy (CAO), drawing on the concept of spatial occupancy previously observed in aerial traffic monitoring [2]. Rather than relying on discrete object detection within fixed geometric boundaries, the proposed framework employs YOLOv8 semantic segmentation to dynamically isolate the actual drivable road surface. By computing the pixel ratio of vehicle footprints relative to the dynamically segmented roadway, this method measures spatial saturation continuously. This approach reduces dependency on rigid ROIs and decreases susceptibility to perspective distortion or minor camera displacements, thereby offering a more robust state estimation alternative for dense, mixed traffic conditions.

3 Methodology

The methodological framework of this study is organized into three sequential subsections. The first subsection describes the overall system architecture, illustrating the closed-loop co-simulation workflow between the GAMA-based Digital Twin platform and the signal control module. The second subsection introduces the YOLOv8 semantic segmentation pipeline for computer vision data extraction, detailing how perspective information is utilized to estimate the Continuous

Area Occupancy (CAO) ratio from virtual camera streams. The third subsection formulates the adaptive C-MP optimization model, which dynamically recalibrates the signal cycle length and green time splits under strict technical and safety constraints.

3.1 System Architecture

To realize the Digital Twin model supporting adaptive signal control for mixed traffic, the operational workflow is designed as a closed-loop framework consisting of eight sequential steps across two core layers: Traffic Simulation and the Signal Control Algorithm. The architectural interaction and data exchange between these layers are illustrated in Figure 1.

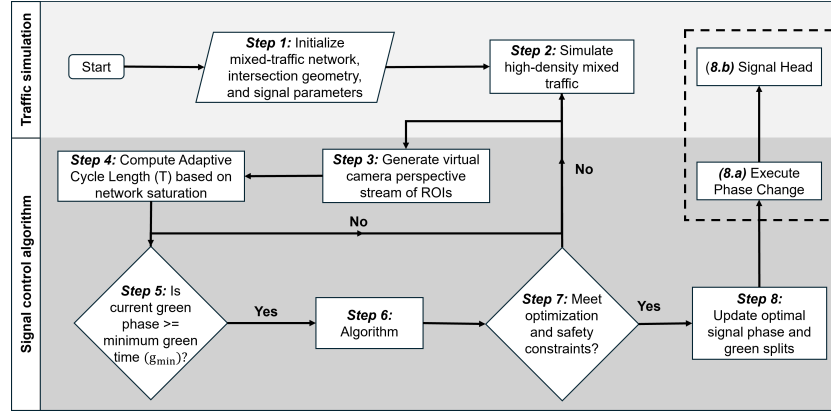


Fig. 1. Systematic flowchart of the spatial-occupancy adaptive signal control architecture across traffic simulation and control layers.

Step 1. The system extracts base map data from the OpenStreetMap (OSM) platform for the designated case study area in Can Tho City, Vietnam. QGIS is subsequently used to standardize the geometric dimensions of the roadways according to real-world scales, precisely mapping the coordinate network, adjacent buildings, intersection geometries, and the spatial distribution of signal heads. This spatial dataset is then integrated into the GAMA multi-agent simulation environment to establish a real-time visual rendering pipeline.

Step 2. Traffic characteristics in a developing urban area such as Can Tho are highly chaotic, characterized by non-lane-based, free-flowing movements. Based on field data collected via surveillance cameras at the target intersection during peak hours, the vehicle composition is configured with a dominant motorcycle ratio: 84% motorcycles, 11% passenger cars, and 5% trucks. To stress-test the system's capacity boundary and resilience, three incremental traffic demand levels are applied: low (400 vph), medium (900 vph), and high (1400 vph). Since

wide-area camera data across the entire network is unavailable, turning ratio distributions are modeled via two predefined routing configurations: *Balanced Turning Ratios*, where flows are distributed evenly with 70% through, 15% left-turn, and 15% right-turn movements; and *Unbalanced Turning Ratios*, which simulate severe asymmetric directional pressures with 50% through, 35% left-turn, and 15% right-turn movements.

Step 3. Virtual camera nodes are deployed at each intersection approach within the simulation environment to capture perspective visual streams. The Regions of Interest are configured dynamically, where the spatial length of each ROI is determined based on the actual physical distance from the adjacent upstream intersection to the stop line of the current approach.

Step 4. From the visual streams obtained in Step 3, the Continuous Area Occupancy (CAO) ratio is extracted to accurately reflect the dynamic saturation state of the road surface, replacing discrete vehicle counts. This metric is subsequently used to determine the optimal adaptive cycle length (T) for the intersection at the current control interval. This cycle length then serves as a direct input to the phase allocation algorithm in Step 6.

Step 5. The system evaluates the elapsed time of the active green phase against a predefined safety threshold of $g_{\min} = 10$ seconds, established based on standardized urban traffic safety regulations [3] to guarantee sufficient pedestrian crossing time and prevent vehicles from entering the dilemma zone. If the condition is not satisfied, the system rejects phase termination and reverts to Step 2 to maintain the current simulation state. If the condition is satisfied, the process advances to Step 6.

Step 6. The enhanced Cycle-Based Max Pressure (C-MP) algorithm is activated to generate the optimal signal timing plan. The core contribution lies in the reformulation of the pressure weight (w): instead of counting discrete vehicles as in the standard C-MP formulation (Eq. (2)), the system incorporates the continuous spatial occupancy ratio ϕ (Eq. (3)) derived from the YOLOv8 semantic segmentation model. Based on these spatial pressure weights, Eq. (3), Eq. (4), and Eq. (5) are solved sequentially to yield the optimal green splits and phase allocations.

Step 7. The green splits derived from Step 6 are validated against two system constraints (Eq. (6) and Eq. (7)): a maximum green time ceiling ($g_{\max}$) to prevent starvation on opposing approaches, and a boundary constraint ensuring that the sum of all phase durations equals the adaptive cycle length T computed in Step 4. If either constraint is violated, the proposed configuration is discarded to prevent network destabilization, and the loop reverts to Step 2. If both constraints are satisfied, the approved configuration is forwarded to the actuation layer.

Step 8. Upon clearing all constraint filters, the system applies Eq. (8) to convert the optimal phase ratios into precise green durations in seconds. Execution proceeds through two parallel actions: (8.a) *Execute Phase Change* issues the phase transition commands and transmits the control packets to the signal controller; and (8.b) *Signal Head Update* dynamically renders the corresponding signal states (Green/Yellow/Red) onto the physical signal structures within the

GAMA simulation environment, completing a full closed-loop feedback cycle of the traffic Digital Twin.

3.2 Area-Based Occupancy Pressure Weight Modeling

Area-Based Occupancy Detection Operator. At each traffic link n , the spatial road occupancy ratio $\phi_n(k)$ within control cycle k is defined as:

$$\phi_n(k) = \frac{\sum_{i=1}^{N_n(k)} \text{Area}(V_{i,n}(k) \cap Z_n)}{\text{Area}(Z_n)} \quad (1)$$

Where:

- $\phi_n(k)$ is the spatial occupancy ratio of link n at control cycle k , bounded by $0 \leq \phi_n(k) \leq 1$;
- $V_{i,n}(k)$ denotes the bounding polygon representing the spatial footprint of the i -th vehicle on link n ;
- Z_n is the designated detection zone on link n ;
- $\text{Area}(\cdot)$ denotes the geometric area computation operator;
- $N_n(k)$ is the total number of vehicle entities detected within link n at control cycle k .

Enhanced Pressure Weighting Based on Area Occupancy. The classic Max Pressure control model [12] relies on the physical vehicle count $x_l(k)$ to determine the pressure weight for each traffic movement:

$$w_{l,m}(k) = x_l(k) - \sum_{p \in \mathcal{O}(m)} R_{m,p} x_p(k) \quad (2)$$

To overcome the limitations of traditional vehicle counting under non-lane-based mixed traffic conditions, this study proposes an enhanced pressure weight formulation based on the spatial road occupancy ratio $\phi(k)$. The area-based pressure weight ($w_{l,m}^{\text{area}}$) is defined as follows:

$$w_{l,m}^{\text{area}}(k) = \phi_l(k) - \sum_{p \in \mathcal{O}(m)} R_{m,p} \phi_p(k) \quad (3)$$

Where:

- $w_{l,m}(k)$ and $w_{l,m}^{\text{area}}(k)$ represent the original pressure weight and the enhanced area-based pressure weight, respectively, from inbound link l to outbound link m within control cycle k ;
- $x_l(k)$, $x_p(k)$ are the vehicle counts on upstream link l and downstream link p ;
- $\phi_l(k)$, $\phi_p(k)$ denote the spatial road occupancy ratios on link l and link p , extracted automatically from the computer vision model;
- $\mathcal{O}(m)$ is the set of immediate downstream links connected to link m ;
- $R_{m,p}$ is the turning movement probability from link m to link p , derived from the routing matrix $\mathbf{R}$.

Signal Timing Allocation Optimization. Based on the proposed area-occupancy weights, the composite spatial pressure of signal phase S within control cycle k is given by:

$$\gamma_S(k) = \sum_{(l,m) \in \mathcal{M}_S} C_{l,m} w_{l,m}^{\text{area}}(k) \quad (4)$$

Where:

- $\gamma_S(k)$ is the composite pressure of signal phase S within control cycle k ;
- $\mathcal{M}_S$ denotes the set of non-conflicting traffic movements permitted during phase S ;
- $C_{l,m}$ is the saturation flow coefficient of movement (l, m) .

Following the C-MP framework [3], the intersection controller solves a linear optimization problem to maximize the total pressure cleared across the intersection, yielding the optimal green time allocation ratio $\lambda_S^*(k)$ for the subsequent cycle:

$$\lambda_S^*(k) = \max_{\lambda} \sum_{S \in \mathcal{I}} \lambda_S(k) \gamma_S(k) \quad (5)$$

The optimization is subject to the following technical and safety constraints:

$$\sum_{S \in \mathcal{I}} \lambda_S(k) = 1 - \frac{L}{\tau} \quad (6)$$

$$\lambda_S(k) \geq \kappa_S, \quad \forall S \in \mathcal{I} \quad (7)$$

Where:

- $\lambda_S(k)$ is the green time ratio allocated to phase S within control cycle k ;
- $\mathcal{I}$ denotes the set of all signal phases at the intersection;
- L is the total lost time per cycle due to phase transitions;
- τ is the fixed signal cycle length;
- κ_S is the minimum green time ratio for phase S , ensuring pedestrian safety and standard intersection operations.

Given the optimal solution $\lambda_S^*(k)$ from Eq. (5), the actual green duration in seconds allocated to phase S within cycle k is:

$$g_S(k) = \lambda_S^*(k) \times \tau \quad (8)$$

Where $g_S(k)$ is the green signal duration for phase S within control cycle k (unit: seconds).

4 Experimental Framework

To implement a safe sandbox testing environment for the enhanced C-MP signal control algorithm, this study adopts the Agent-Based Modeling (ABM) approach, which enables traffic entities to be modeled as autonomous agents possessing individual behavioral attributes and the capacity for non-linear interactions with the environment. The open-source GAMA multi-agent simulation platform [11,8] is selected as the microscopic simulation core for the proposed Digital Twin system.

4.1 Simulation Set-up

The microscopic simulation environment is constructed on the GAMA multi-agent simulation platform, utilizing spatial data from OpenStreetMap (OSM) to accurately digitize the geometric layout of the target intersection in Can Tho City. To ensure high spatial fidelity prior to importation, the open-source QGIS software is used to process, align, and refine the raw geodata. The resulting simulation layout, which captures the complex intersection geometry, is visualized in Fig. 2.

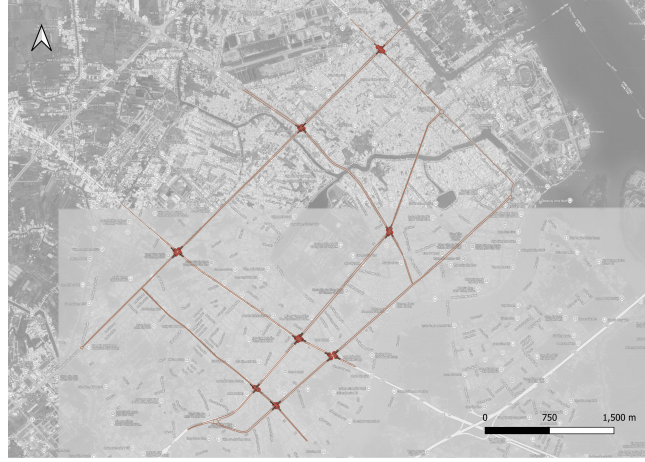


Fig. 2. Spatial layout of the target intersection processed via QGIS and imported into the GAMA simulation platform.

To capture the non-lane-based swarming behavior of mixed traffic flows, the simulation integrates customized Virtual Detection Zones (VDZs) in place of traditional physical lanes. Within this ABM framework, the environment comprising the road network, traffic signal controllers, and detection zones dynamically interacts with heterogeneous vehicle agents (motorcycles, passenger cars, and trucks). Operating as a closed-loop Digital Twin system, real-time Continuous Area Occupancy (CAO) values enable the C-MP algorithm to update signal timings at each control cycle. The GIS-based modular structure ensures transferability, providing a scalable template for district-level adaptive signal control without requiring modification of the core mathematical logic.

4.2 Evaluation Scenarios

To validate the operational performance of the proposed CAO-based C-MP controller, a comparative simulation framework is established within the GAMA multi-agent simulation platform. The proposed model is benchmarked against

two baseline strategies: (1) a conventional fixed-time signal control plan, and (2) a standard vehicle-counting-based C-MP algorithm. Two routing configurations are designed to evaluate structural resilience under varying directional pressures: Balanced Turning Ratios, where flows are distributed evenly among left-turn, through, and right-turn movements, and Unbalanced Turning Ratios, where heavy asymmetric directional flows simulate peak-hour tidal traffic patterns. Three incremental traffic demand levels 400, 900, and 1400 vph are applied to stress-test the intersection across different Levels of Service (LOS). The vehicle composition is configured as a motorcycle-dominated environment to capture realistic swarming and filtering behaviors.

Table 1. Evaluation Scenarios

Scenarios	Concept	Traffic Movements	Traffic Demand
Baseline	Fixed-time signal control with pre-determined phase durations, serving as the performance reference.	Standard 8-movement intersection structure: – <i>Through:</i> EBT, WBT, SBT, NBT – <i>Left-turn:</i> EBL, WBL, SBL, NBL	The vehicle composition at each network entry is configured as 85% motorcycles, 12% passenger cars, and 3% trucks, under three demand levels: – 400 vph – 900 vph – 1400 vph
Standard Cycle-Based Max-Pressure	Standard C-MP algorithm. Signal optimization is based on discrete vehicle counts within the detection zone, with calibrated conversion weights for motorcycles.	<i>Distributed into two patterns:</i> – Balanced: 70% through, 15% left, 15% right. – Unbalanced: 50% through, 35% left, 15% right.	
Proposed Cycle-Based Max-Pressure	Proposed enhanced C-MP model. Phase durations are optimized strictly based on the Continuous Area Occupancy (CAO) ratio $\phi_n(k)$ of the mixed traffic stream.		

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